

CN PA-1 Design Document

Group 31

1. I added a programmable packet dropper in the E1000 transmit path. The driver listens for a special control packet (UDP to 100.100.100.100:100 with 5-byte payload {a,b,c,d,rate}) and updates global state: the target IP and drop rate.

At the start of `e1000_xmit_frame()`, the driver checks outgoing IPv4 packets:

- If it's the control packet → parse and store {target, rate}.
- If it's destined to the target → randomly drop with probability rate%. Dropped packets are freed and counted in `tx_dropped`.

Key changes:

- Added globals (`g_target_ip`, `g_drop_rate`, `g_drop_lock`) and initialized lock in `e1000_probe()`.
- Inserted the control-message parser + probabilistic drop logic at the top of `e1000_xmit_frame()`.

```
1998 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./block 192.168.43.247 0
Blocking: 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./client 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
1998 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$
```

```
completed 80000 packets
completed 81000 packets
completed 82000 packets
completed 83000 packets
completed 84000 packets
completed 85000 packets
completed 86000 packets
completed 87000 packets
completed 88000 packets
completed 89000 packets
completed 90000 packets
completed 91000 packets
completed 92000 packets
completed 93000 packets
completed 94000 packets
completed 95000 packets
completed 96000 packets
completed 97000 packets
completed 98000 packets
completed 99000 packets
completed 100000 packets
completed 101000 packets
101153 packets received in 1672 milliseconds
dhruv@dhruv-Surface-Pro:~/Downloads/usertools-master$
```

2. 1672 ms
3. 1998 segments
4. 1998 segments

```

dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./block 192.168.43.247 1
Blocking: 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
2133 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./client 192.168.43.247
connect: Connection refused
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
2133 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./client 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
2188 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$

```

```

completed 79000 packets
completed 80000 packets
completed 81000 packets
completed 82000 packets
completed 83000 packets
completed 84000 packets
completed 85000 packets
completed 86000 packets
completed 87000 packets
completed 88000 packets
completed 89000 packets
completed 90000 packets
completed 91000 packets
completed 92000 packets
completed 93000 packets
completed 94000 packets
completed 95000 packets
completed 96000 packets
completed 97000 packets
completed 98000 packets
completed 99000 packets
completed 100000 packets
100495 packets received in 2269 milliseconds
dhruv@dhruv-Surface-Pro:~/Downloads/usertools-master$

```

5. 2269 ms
6. 2133 segments
7. 2188 segments

```

2188 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./block 192.168.43.247 5
Blocking: 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
2188 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./client 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
2432 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$

```

```

completed 79000 packets
completed 80000 packets
completed 81000 packets
completed 82000 packets
completed 83000 packets
completed 84000 packets
completed 85000 packets
completed 86000 packets
completed 87000 packets
completed 88000 packets
completed 89000 packets
completed 90000 packets
completed 91000 packets
completed 92000 packets
completed 93000 packets
completed 94000 packets
completed 95000 packets
completed 96000 packets
completed 97000 packets
completed 98000 packets
completed 99000 packets
completed 100000 packets
100518 packets received in 11170 milliseconds
dhruv@dhruv-Surface-Pro:~/Downloads/usertools-master$

```

8. 11170 ms

9. 2188 segments

10. 2432 segments

```

dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
2432 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./block 192.168.43.247 10
Blocking: 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./client 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
3044 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$

```

```

completed 79000 packets
completed 80000 packets
completed 81000 packets
completed 82000 packets
completed 83000 packets
completed 84000 packets
completed 85000 packets
completed 86000 packets
completed 87000 packets
completed 88000 packets
completed 89000 packets
completed 90000 packets
completed 91000 packets
completed 92000 packets
completed 93000 packets
completed 94000 packets
completed 95000 packets
completed 96000 packets
completed 97000 packets
completed 98000 packets
completed 99000 packets
completed 100000 packets
100492 packets received in 61530 milliseconds
dhruv@dhruv-Surface-Pro:~/Downloads/usertools-master$

```

11. 61530 ms

12. 2432 segments

13. 3044 segments

```
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
3044 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./block 192.168.43.247 15
Blocking: 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./client 192.168.43.247
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ netstat -s | grep retransmitted
3947 segments retransmitted
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$
```

```
completed 79000 packets
completed 80000 packets
completed 81000 packets
completed 82000 packets
completed 83000 packets
completed 84000 packets
completed 85000 packets
completed 86000 packets
completed 87000 packets
completed 88000 packets
completed 89000 packets
completed 90000 packets
completed 91000 packets
completed 92000 packets
completed 93000 packets
completed 94000 packets
completed 95000 packets
completed 96000 packets
completed 97000 packets
completed 98000 packets
completed 99000 packets
completed 100000 packets
100617 packets received in 106357 milliseconds
dhruv@dhruv-Surface-Pro:~/Downloads/usertools-master$
```

14. 106357 ms
15. 3044 segments
16. 3947 segments

```
-n1
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ dig +short www.google.com | head
-n1
142.250.193.4
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ./block 142.250.193.4 100
Blocking: 142.250.193.4
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ping -c 3 www.ggoogle.com
ping: www.ggoogle.com: Name or service not known
dhruv@dhruv-Ubuntu-24-04-PC-i440FX-PIIX-1996:~$ ping -c 3 www.google.com
PING www.google.com (142.250.193.4) 56(84) bytes of data.

--- www.google.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2078ms
```

17. No

```
completed 31000 packets
completed 32000 packets
completed 33000 packets
completed 34000 packets
completed 35000 packets
completed 36000 packets
completed 37000 packets
completed 38000 packets
completed 39000 packets
completed 40000 packets
completed 41000 packets
completed 42000 packets
completed 43000 packets
completed 44000 packets
completed 45000 packets
completed 46000 packets
completed 47000 packets
completed 48000 packets
completed 49000 packets
completed 50000 packets
completed 51000 packets
completed 52000 packets
52029 packets received in 2862 milliseconds
dhruv@dhruv-Surface-Pro:~/Downloads/usertools-master$
```

18. Packet size almost halved. Amount of data being sent is the same but packet size doubles from 64 bytes to 128. Larger buffer size changes number of calls needed to drain the stream.

19. Dhruv Jain - 2023199

Shivank Rajput - 2023508

Altamash Hasnain - 2023073

Divyanshi - 2023209

Saksham Arora - 2023466